"How Good Is Your Explanation?": Towards a Standardised Evaluation Approach for Diverse XAI Methods on Multiple Dimensions of Explainability

Aditya Bhattacharya
aditya.bhattacharya@kuleuven.be
KU Leuven
Leuven, Belgium

Katrien Verbert
katrien.verbert@kuleuven.be
KU Leuven
Leuven, Belgium

ABSTRACT

Artificial Intelligence (AI) systems involve diverse components, such as data, models, users and predicted outcomes. To elucidate these different aspects of AI systems, multifaceted explanations that combine diverse explainable AI (XAI) methods are beneficial. However, popularly adopted user-centric XAI evaluation methods do not measure these explanations across the different components of the system. In this position paper, we advocate for an approach tailored to evaluate XAI methods considering the diverse dimensions of explainability within AI systems using a normalised scale. We argue that the prevalent user-centric evaluation methods fall short of facilitating meaningful comparisons across different types of XAI methodologies. Moreover, we discuss the potential advantages of adopting a standardised approach, which would enable comprehensive evaluations of explainability across systems. By considering various dimensions of explainability, such as data, model, predictions, and target users, a standardised evaluation approach promises to facilitate both inter-system and intra-system comparisons for user-centric AI systems.

CCS CONCEPTS

• **Human-centered computing** → **Human computer interaction (HCI)**; **Interaction design**; • **Computing methodologies** → **Machine learning**.

KEYWORDS

Explainable AI, XAI, Explainable AI Evaluation

ACM Reference Format:
Aditya Bhattacharya and Katrien Verbert. 2024. "How Good Is Your Explanation?": Towards a Standardised Evaluation Approach for Diverse XAI Methods on Multiple Dimensions of Explainability. In *Adjunct Proceedings of the 32nd ACM Conference on User Modeling, Adaptation and Personalization (UMAP Adjunct '24), July 01–04, 2024, Cagliari, Italy.* ACM, New York, NY, USA, 3 pages. https://doi.org/10.1145/3631700.3664911

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than the author(s) must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from permissions@acm.org.
UMAP Adjunct '24, July 01–04, 2024, Cagliari, Italy
© 2024 Copyright held by the owner/author(s). Publication rights licensed to ACM.
ACM ISBN 979-8-4007-0466-6/24/07
https://doi.org/10.1145/3631700.3664911

1 INTRODUCTION

Artificial Intelligence (AI) systems constitute multiple components, such as the data, models, predictions and the users who will use such systems. Explainable AI (XAI) methods are designed to elucidate these different components of "black-box" AI systems [1, 8]. The act of offering explanations throughout the distinct components of an AI system is also referred to as *dimensions of explainability* [3, 20]. Recent works have demonstrated the benefits of combining different types of XAI methods in multifaceted explanations, particularly for non-expert users in AI [5–7].

Additionally, it has been argued that the *No Free Lunch* theorem in Machine Learning [11] (i.e., there is no one-size fits all algorithm for all tasks or datasets) also applies to XAI, as specific XAI methods can only explain certain specific dimensions of explainability [7]. Thus, it is essential to include multiple types of XAI methods for a holistic explanation of AI systems. For example, an XAI system with both feature importance explanations and counterfactual explanations can provide more holistic explanations rather than a system using only one of these methods. However, these methods elucidate completely different dimensions of explainability. The former tries to explain the importance of certain factors considered by the model for generating predictions (i.e., model-centric explanations), whereas the latter provides various conditions for obtaining a different prediction (i.e., outcome-centric explanations). Therefore, a specific XAI method can elucidate a specific dimension of explainability.

The process of user-centric evaluation for XAI predominantly involves user-reported scores for metrics such as *understandability*, *trust*, *actionability*, *stability*, *usefulness*, and etc., from the participants involved in user studies [13, 14, 17, 21]. Despite the significance of such methods, their implications can be limited for evaluating an XAI system as they do not consider different explainability dimensions other than the user perspective. Moreover, it could be difficult to compare diverse XAI methods only based on user perspectives due to certain limitations, such as users agreeing to misleading explanations even when the system makes incorrect predictions [9, 18, 19].

To overcome these challenges, researchers have also adopted other approaches that provide an objective evaluation of XAI methods, such as task-driven approaches [10, 16], algorithmic evaluation metrics [2], and model-specific measurement metrics similar to Quantus [12]. However, estimating the individual impact of a particular XAI method becomes challenging in the presence of multiple combined XAI methods within a system. Thus, despite various objective and subjective metrics for measuring XAI methods [15],

comparing diverse methods considering the different dimensions of explainability remains an unsolved problem [5]. To enable comparative evaluation studies, there is an evident need for a normalised XAI evaluation approach that considers both subjective and objective metrics across the various dimensions of explainability. This necessity raises an essential open research question: "*How can we compare XAI methods elucidating different dimensions of explainability when used in XAI systems?*"

In this position paper, we discuss the necessity of a standardised approach for evaluating diverse explainability methods used within XAI systems, considering the diverse explainability dimensions. We also discuss the potential advantages of adopting a standardised approach, which would enable comprehensive evaluations of explainability across systems. We believe that existence of such an approach can be extremely beneficial for the Human Centred XAI (HCXAI) community for evaluating XAI systems.

2 THE NEEDS FOR A STANDARDISED EVALUATION APPROACH FOR EXPLAINABLE AI SYSTEMS

A standardised evaluation methodology for assessing the explainability of diverse XAI methods within AI systems could effectively mitigate challenges associated with cross-dimensional comparisons. Such an approach should integrate both objective and subjective evaluation metrics, standardised to a consistent scale. By normalising evaluation scores across dimensions, such an approach could facilitate meaningful comparisons of the efficacy of various XAI methods in elucidating the multifaceted components of AI systems.

We highlight the following needs that could be fulfilled by a standardised evaluation approach for XAI systems:

- *Holistic Measurement*: This metric should provide a standardised and holistic measure of the effectiveness of different XAI methods in elucidating the multiple components of AI systems. This addresses the need for a comprehensive evaluation that goes beyond individual metrics.
- *Flexibility*: The standardised approach should offer flexibility in incorporating existing evaluation metrics across individual dimensions of explainability. This should encompass objective measures assessing training data quality, model performance, and considerations for prediction uncertainty alongside user-centric evaluations such as trustworthiness, understandability, and others.
- *Model-Agnostic Property*: This normalised evaluation approach should be model-agnostic, i.e., it could be applied to evaluate any XAI method or AI models. A model-agnostic evaluation approach could broaden its applicability to different application domains and diverse AI systems.
- *Intra-System and Inter-System Comparison*: The normalised approach should be used for comparing two or more XAI systems. It should also allow individual XAI methods across the different explainability dimensions. However, the main goal of such an approach is to compare different XAI methods within a system, using both subjective and objective measures.

3 SUMMARY

To summarise, in this position paper, we discuss the needs for an approach designed to assess different dimensions of explainability of diverse XAI methods within XAI systems. We also discuss the benefits and opportunities for leveraging this approach for evaluating XAI systems. We eagerly seek feedback from the workshop participants to refine and operationalise these ideas into a robust evaluation framework.

ACKNOWLEDGMENTS

We thank Maxwell Szymanski and Robin De Croon for their valuable feedback on this research. This research was supported by Research Foundation–Flanders (FWO grants G0A3319N, G0A4923N and G067721N) and KU Leuven Internal Funds (grant C14/21/072) [4].

REFERENCES

[1] Amina Adadi and Mohammed Berrada. 2018. Peeking Inside the Black-Box: A Survey on Explainable Artificial Intelligence (XAI). *IEEE Access* 6 (2018), 52138–52160. https://doi.org/10.1109/ACCESS.2018.2870052

[2] Chirag Agarwal, Satyapriya Krishna, Eshika Saxena, Martin Pawelczyk, Nari Johnson, Isha Puri, Marinka Zitnik, and Himabindu Lakkaraju. 2022. OpenXAI: Towards a Transparent Evaluation of Model Explanations. In *Thirty-sixth Conference on Neural Information Processing Systems Datasets and Benchmarks Track*. https://openreview.net/forum?id=MU2495w47rz

[3] Aditya Bhattacharya. 2022. Applied Machine Learning Explainability Techniques. In *Applied Machine Learning Explainability Techniques*. Packt Publishing, Birmingham, UK. https://www.packtpub.com/product/applied-machine-learning-explainability-techniques/9781803246154

[4] Aditya Bhattacharya. 2024. Towards Directive Explanations: Crafting Explainable AI Systems for Actionable Human-AI Interactions. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '24)* (Honolulu, HI, USA) *(CHI EA '24)*. ACM, New York, NY, USA, 6. https://doi.org/10.1145/3613905.3638177

[5] Aditya Bhattacharya, Jeroen Ooge, Gregor Stiglic, and Katrien Verbert. 2023. Directive Explanations for Monitoring the Risk of Diabetes Onset: Introducing Directive Data-Centric Explanations and Combinations to Support What-If Explorations. In *Proceedings of the 28th International Conference on Intelligent User Interfaces* (Sydney, NSW, Australia) *(IUI '23)*. Association for Computing Machinery, New York, NY, USA, 204–219. https://doi.org/10.1145/3581641.3584075

[6] Aditya Bhattacharya, Simone Stumpf, Lucija Gosak, Gregor Stiglic, and Katrien Verbert. 2023. Lessons Learned from EXMOS User Studies: A Technical Report Summarizing Key Takeaways from User Studies Conducted to Evaluate The EXMOS Platform. arXiv:2310.02063 [cs.LG]

[7] Aditya Bhattacharya, Simone Stumpf, Lucija Gosak, Gregor Stiglic, and Katrien Verbert. 2024. EXMOS: Explanatory Model Steering Through Multifaceted Explanations and Data Configurations. In *Proceedings of the CHI Conference on Human Factors in Computing Systems (CHI '24)* (Honolulu, HI, USA) *(CHI '24)*. ACM, New York, NY, USA, 27. https://doi.org/10.1145/3613904.3642106

[8] Carrie J. Cai, Emily Reif, Narayan Hegde, Jason Hipp, Been Kim, Daniel Smilkov, Martin Wattenberg, Fernanda Viegas, Greg S. Corrado, Martin C. Stumpe, and Michael Terry. 2019. Human-Centered Tools for Coping with Imperfect Algorithms during Medical Decision-Making. https://doi.org/10.48550/ARXIV.1902.02960

[9] Michael Chromik, Malin Eiband, Felicitas Buchner, Adrian Krüger, and Andreas Butz. 2021. I Think I Get Your Point, AI! The Illusion of Explanatory Depth in Explainable AI. In *26th International Conference on Intelligent User Interfaces* (College Station, TX, USA) *(IUI '21)*. Association for Computing Machinery, New York, NY, USA, 307–317. https://doi.org/10.1145/3397481.3450644

[10] Finale Doshi-Velez and Been Kim. 2017. Towards A Rigorous Science of Interpretable Machine Learning. arXiv:1702.08608 [stat.ML]

[11] Micah Goldblum, Marc Finzi, Keefer Rowan, and Andrew Gordon Wilson. 2023. The No Free Lunch Theorem, Kolmogorov Complexity, and the Role of Inductive Biases in Machine Learning. arXiv:2304.05366 [cs.LG]

[12] Anna Hedström, Leander Weber, Daniel Krakowczyk, Dilyara Bareeva, Franz Motzkus, Wojciech Samek, Sebastian Lapuschkin, and Marina Marina M.-C. Höhne. 2023. Quantus: An Explainable AI Toolkit for Responsible Evaluation of Neural Network Explanations and Beyond. *Journal of Machine Learning Research* 24, 34 (2023), 1–11. http://jmlr.org/papers/v24/22-0142.html

[13] Robert R. Hoffman, Shane T. Mueller, Gary Klein, and Jordan Litman. 2019. Metrics for Explainable AI: Challenges and Prospects. arXiv:1812.04608 [cs.AI]

[14] Q. Vera Liao, Yunfeng Zhang, Ronny Luss, Finale Doshi-Velez, and Amit Dhurandhar. 2022. Connecting Algorithmic Research and Usage Contexts: A Perspective of Contextualized Evaluation for Explainable AI. arXiv:2206.10847 [cs.AI]

[15] Sina Mohseni, Niloofar Zarei, and Eric D. Ragan. 2021. A Multidisciplinary Survey and Framework for Design and Evaluation of Explainable AI Systems. *ACM Trans. Interact. Intell. Syst.* 11, 3–4, Article 24 (sep 2021), 45 pages. https://doi.org/10.1145/3387166

[16] Katelyn Morrison, Mayank Jain, Jessica Hammer, and Adam Perer. 2023. Eye into AI: Evaluating the Interpretability of Explainable AI Techniques through a Game with a Purpose. *Proc. ACM Hum.-Comput. Interact.* 7, CSCW2, Article 273 (oct 2023), 22 pages. https://doi.org/10.1145/3610064

[17] Meike Nauta, Jan Trienes, Shreyasi Pathak, Elisa Nguyen, Michelle Peters, Yasmin Schmitt, Jörg Schlötterer, Maurice van Keulen, and Christin Seifert. 2023. From Anecdotal Evidence to Quantitative Evaluation Methods: A Systematic Review on Evaluating Explainable AI. *ACM Comput. Surv.* 55, 13s, Article 295 (jul 2023), 42 pages. https://doi.org/10.1145/3583558

[18] Forough Poursabzi-Sangdeh, Daniel G Goldstein, Jake M Hofman, Jennifer Wortman Wortman Vaughan, and Hanna Wallach. 2021. Manipulating and Measuring Model Interpretability. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems* (<conf-loc>, <city>Yokohama</city>, <country>Japan</country>, </conf-loc>) *(CHI '21)*. Association for Computing Machinery, New York, NY, USA, Article 237, 52 pages. https://doi.org/10.1145/3411764.3445315

[19] J. Schneider, C. Meske, and M. Vlachos. 2024. Deceptive XAI: Typology, Creation and Detection. *SN COMPUT. SCI.* 5 (2024), 81. https://doi.org/10.1007/s42979-023-02401-z

[20] Kacper Sokol and Peter Flach. 2020. Explainability fact sheets: a framework for systematic assessment of explainable approaches. In *Proceedings of the 2020 Conference on Fairness, Accountability, and Transparency* (Barcelona, Spain) *(FAT* '20)*. Association for Computing Machinery, New York, NY, USA, 56–67. https://doi.org/10.1145/3351095.3372870

[21] Nava Tintarev and Judith Masthoff. 2011. *Designing and Evaluating Explanations for Recommender Systems.* 479–510. https://doi.org/10.1007/978-0-387-85820-3_15